

Temperature Kills Models, Not Answers: Decoding Robustness Is a Model Property

Anonymous ACL submission

Abstract

Truncation samplers such as top- p , min- p , and top- $n\sigma$ exist to prevent the accuracy collapse of high-temperature sampling. We ask whether the collapse itself is universal. Using a single controlled pipeline, we evaluate 9 instruction-tuned models from 5 model families (1.7B to 12B parameters), 4 quantization levels, 8 sampler arms, and 2 tasks, totaling roughly 200,000 generations. At deployment temperatures ($T \leq 1.3$), sampler choice barely matters: outside one family, the per-model accuracy spread across samplers is 1.3 to 10.0 percentage points (pp), and even that overstates the true differences. The exception is the Llama family. Under plain temperature sampling at $T=1.3$, Llama-3.1-8B loses 34 pp (GSM8K) and 41 pp (MMLU-Pro), Llama-3.2-3B collapses the same way, and Llama-3-8B loses 17 pp on MMLU-Pro. Every model outside the family stays within 8 pp, including Mistral-7B-v0.3, which is older than Llama-3.1, and Qwen3-1.7B, whose low-temperature accuracy is close to Llama-3.1's. The collapse is a termination failure: generations derail midway and never reach an answer, while accuracy conditional on producing a well-formed answer stays at the greedy level in every cell (97% for Llama-3.1-8B on GSM8K at $T=1.3$). A temperature ladder up to $T=2.0$ locates each model's cliff and shows that truncation samplers only begin to matter past it. A logit probe and a forced-token perturbation experiment trace the fragility to a self-reinforcing instability: one off-distribution token in context multiplies the fragile model's chance of drawing the next one. Practitioners tuning a current model can leave the sampler alone.

1 Introduction

Most deployed language models generate text by sampling. At each step the model assigns a probability to every candidate next token; the decoder either takes the most probable one (greedy decod-

ing) or draws at random from the distribution. Systems sample because greedy decoding returns one fixed continuation per prompt, and in open-ended text that continuation tends to be repetitive (Holtzman et al., 2020). A temperature parameter scales the randomness of the draw: raising it moves probability from the model's preferred tokens toward rare ones.

The rare tokens are where sampling goes wrong. A model's probability estimates are least reliable in the tail of its distribution, so a high-temperature draw can select a token that fits the context poorly (Holtzman et al., 2020; Hewitt et al., 2022). The selected token then becomes context for every later step, and the continuation can degenerate into incoherent text. Truncation samplers exist to prevent this degeneration: they delete the tail of the distribution before the draw and renormalize what remains. The rules differ in where they cut the tail; top- k , nucleus (top- p), min- p , and top- $n\sigma$ sampling are the four we test, and Section 2 describes each (Fan et al., 2018; Holtzman et al., 2020; Nguyen et al., 2025; Tang et al., 2024). All of these were developed for open-ended generation, such as story writing and dialogue, where output variety is the goal (Garces Arias et al., 2024).

The same samplers are now evaluated on tasks with one checkable answer. Deployed assistants keep sampling on for every request, so math word problems and multiple-choice exams are also decoded at nonzero temperature (Song et al., 2025; Renze and Guven, 2024). On such tasks a decoding method is scored by accuracy: each generation is right or wrong, where open-ended generation needs a human or model judge (Garces Arias et al., 2024). Min- p , for example, reports accuracy gains over top- p on GSM8K at high temperature (Nguyen et al., 2025).

Those accuracy claims drew a critique. Schaeffer et al. (2025) asked whether any one sampler beats the others once statistical controls are

in place. Whichever sampler wins that comparison, the comparison itself takes the collapse for granted: it assumes that raising the temperature eventually ruins every model’s accuracy, and asks only which sampler limits the damage. We test the premise: **is the collapse itself a property of decoding, or of the model?**

Across nine instruction-tuned models from five families, run through one controlled pipeline, the answer is: of the model. The six models from outside the Llama family hold their accuracy at deployment temperatures under every sampler we test, staying within 8 points of their low-temperature scores. Every larger loss belongs to a Llama: Llama-3.1-8B and Llama-3.2-3B lose 34 to 41 accuracy points under plain temperature sampling, and Llama-3-8B loses 17 on the longer task. The collapsing generations derail into token noise before reaching an answer; the answers these models do complete stay as accurate as greedy decoding. Sampler choice, the variable the truncation literature tunes, is worth at most a few points anywhere below each model’s collapse temperature.

Our contributions:

- **A single-pipeline factorial study.** Seven instruction-tuned models, four GGUF quantization levels (model weights stored at reduced precision) plus a BF16 anchor, six decoding methods plus two ablation arms that swap the order in which temperature and truncation are applied, $T \in \{0.7, 1.0, 1.3\}$, on GSM8K and MMLU-Pro with $N=50$ items and $K=3$ repetitions, about 204,000 generations on commodity GPUs, with item-clustered bootstrap confidence intervals throughout, plus a follow-up temperature-only cell on two more Llama generations.
- **Robustness is model-bound, and the fragility runs in one family.** Every collapse we observe is a Llama. In the main grid Llama-3.1-8B loses 34 and 41 pp at $T=1.3$ while Mistral-7B, Qwen2.5-7B, Qwen3-1.7B/4B/8B, and Gemma-3-12B all hold within 8 pp; in the follow-up cell Llama-3.2-3B collapses the same way and Llama-3-8B falls in between (Section 4.1). On the robust models the sampler is worth at most a few points, and even that is an overestimate: taking the max minus the min of noisy accuracy estimates inflates the range with the

noise itself, so the true sampler differences are smaller still.

- **The collapse is a termination failure.** Conditional on emitting a well-formed final answer, accuracy is flat in temperature everywhere. What varies is the probability of derailling into unparseable text mid-generation (Section 4.3).
- **A cliff map from $T=1.3$ to 2.0.** Extending the temperature axis shows that every model collapses somewhere; what distinguishes models is where. On GSM8K, Llama collapses by $T=1.5$, Qwen2.5 by 1.7, and Gemma-3 by 2.0. From $T=1.7$ the separation is statistically clear: the confidence intervals on the three models’ drops do not overlap one another. Truncation samplers shift the cliff right in order of tail-cutting strength ($\text{top-}n\sigma \gg \text{min-}p \gg \text{top-}p$). The gains the sampler papers report are real, but they appear past the cliff, above deployment temperatures.
- **Supporting results.** Quantization from Q8 down to Q3 causes no collapse on any model. It also does not change which sampler ranks best. Its accuracy penalty grows as models shrink (−9 pp on Qwen3-1.7B, null at 7B and above). A logit probe, a length-censoring analysis, and a forced-token experiment (Section 4.6) locate the fragility: every model draws rare off-distribution tokens at the same per-token rate, but one injected tail token multiplies Llama’s subsequent escape hazard 2.3-fold while every robust model absorbs it; the size of that multiplier ranks with the collapse across models ($\rho = +0.86$).

2 Related work

Truncation samplers. Top- k sampling draws from only the k most probable tokens; it was introduced for story generation, where likelihood-maximizing decoding produced generic and repetitive text (Fan et al., 2018). Holtzman et al. (2020) showed that a fixed k fits poorly because the next-token distribution is sometimes peaked and sometimes flat, and proposed nucleus sampling, which keeps the smallest set of tokens whose probabilities sum to a target p . Min- p keeps only the tokens whose probability is at least a set fraction of the top token’s, so the cut is tight where the model is confident and loose where it is uncertain (Nguyen

et al., 2025). Top- $n\sigma$ cuts in logit space instead, keeping tokens within n standard deviations of the top logit; temperature rescales all logits equally, so the surviving set does not change with temperature (Tang et al., 2024). Further refinements of the cut exist, among them typical sampling and epsilon and eta sampling (Meister et al., 2023; Hewitt et al., 2022); we test the four rules above (Section 3). All of these rules target the same failure: incoherent text drawn from the tail at high temperature. Welleck et al. (2020) proved the worst case of that failure, showing that standard decoders can place probability on generations that never terminate. In contrast to the sampler papers, which compare truncation rules against each other, we test whether the failure they all guard against occurs at deployment temperatures at all. Outside the Llama family it does not, and on the Llamas it takes the non-terminating form: generations run to the token cap without reaching an answer (Section 4.3).

Reproducibility of sampler gains. Schaeffer et al. (2025) reanalyzed min- p on nine models and 31 temperatures, giving every sampler an equal number of hyperparameter settings, and found its reported gains “largely indistinguishable” from the other samplers’. Part of the gains traced to the order in which temperature and truncation are applied. The order matters because the truncation threshold is computed on whichever distribution the rule sees: applied before temperature, the rule cuts the model’s original, sharp distribution; applied after, it cuts one that temperature has already flattened, so the same nominal setting keeps a different set of tokens. We adopt both lessons: we report statistical uncertainty throughout and run the two chain orders as explicit arms (Section 3). That critique pooled its models together, because its question was about the sampler. Our question is about the model, so we keep every model separate and ask whether the collapse happens on each one. Garces Arias et al. (2024) is the closest systematic sweep of decoding hyperparameters, but it scores open-ended generation with quality and diversity metrics, while our tasks have one checkable answer. Song et al. (2025) documents per-model gaps between greedy decoding and sampling on reasoning benchmarks and treats the gaps as evaluation noise to report. Unlike Song et al. (2025), we treat the gap as the object of study, and we trace it to a termination failure in one model family.

Robustness to decoding randomness. Renze and Guven (2024) sampled temperatures from 0.0 to 1.6 on multiple-choice problem solving and found no statistically significant effect on accuracy between 0.0 and 1.0, across models, prompting techniques, and domains. Our results replicate that null inside the deployment band. But the null is pooled across models, and pooling hides the outlier: our per-model ladder shows the heterogeneity beginning just past $T=1.0$, where Llama-3.1-8B falls while every other model holds (Section 4.1). The nearest study is Grover (2026), which also asks what makes a model robust to decoding randomness and concludes that instruction tuning is the primary determinant. Our data refine that conclusion: all seven of our models are instruction-tuned and one still collapses, so instruction tuning cannot be the whole answer. Four design differences let our grid observe what theirs cannot:

- their sweep stops at $T=1.0$, and the Llama collapse happens between 1.0 and 1.3;
- their robust models score 6 to 30% on synthetic tasks, so there is little accuracy left to lose;
- their instruct-versus-base contrast changes quantization at the same time (4-bit versus full precision), while we sweep quantization as its own axis and find it is not the variable;
- they test no truncation samplers.

Fastowski et al. (2025) connect the entropy of a model’s token distributions to its temperature sensitivity on factual question answering and report that larger models are more stable. Our logit probe supports the entropy link (Section 4.6). Size does not decide the outcome in our grid, because Qwen3-1.7B holds where the larger Llama-3.1-8B collapses. We also locate the failure earlier than answer selection: the fragile model’s generations stop terminating, while the answers it does complete stay accurate (Section 4.3).

3 Experimental setup

One pipeline, everything pinned: llama.cpp at a fixed commit (CUDA build), one provider’s imatrix GGUF quantizations with logged SHA256 hashes, official chat templates rendered client-side with the exact prompt string hashed into each record, the native completion API, a pinned sampler chain order (temperature first; the llama.cpp

default of temperature last is included as an ablation arm for top- p and min- p), per-record derived seeds, a deterministic greedy anchor ($K=1$), and $K=3$ repetitions elsewhere.

Tasks. GSM8K (0-shot chain-of-thought, the Llama-eval prompt, numeric exact match) and MMLU-Pro (0-shot chain-of-thought, letter exact match), $N=50$ items each, with parsers faithful to lm-eval-harness. Both tasks grade a generation by parsing a final answer out of the generated text, so the parser decides how a generation that never reaches an answer is graded. The parse path (strict final-answer sentence, flexible fallback, or failure) is logged per record. Parse failure grades as incorrect *and* is reported as a metric; Section 4.3 shows it is a result, not a nuisance. One caveat stated up front: the frozen $N=50$ MMLU-Pro subset is head-of-split, and the split is sorted by category, so the subset is single-category (business). All within-study contrasts hold items fixed; a stratified $N=50$ replication (3–4 items from each of the 14 categories) reproduces the headline result (Section 4.1, Appendix C).

Models and quantization. Llama-3.1-8B, Mistral-7B-v0.3, Qwen2.5-7B, Qwen3-1.7B/4B/8B (thinking off; a thinking-on spot-check in Appendix G), and Gemma-3-12B, all instruction-tuned. Quantization levels Q8_0, Q6_K, Q4_K_M, Q3_K_M, plus a BF16 anchor on Qwen3-4B that matches Q8 within noise.

Inference. Item-clustered nonparametric bootstrap ($B=2000$), paired where the comparison shares items.

Hardware. Generation ran under one pinned llama.cpp commit on two setups: RTX 2080 Ti (11 GB) nodes of a university Slurm cluster (17 of the 21 main-matrix cells) and a single RTX 5070 Ti (16 GB) workstation (the four VRAM-heavy cells and every follow-up run), with per-architecture CUDA builds of the same commit. About 204,000 graded generations in total. Full reproducibility details in Appendix J.

4 Results

4.1 The collapse is model-bound

Table 1 and Figure 1 give the accuracy drop under plain temperature sampling from $T=0.7$ to $T=1.3$, pooled over quantization levels.

The models do not split cleanly into robust and fragile. The Qwen3 family is a tight null, Qwen2.5, Gemma, and Mistral drop 2 to 8 pp with

Table 1: Pure-temperature accuracy drop $T0.7 \rightarrow T1.3$ (pp, 95% item-clustered bootstrap CIs), pooled over quantization levels.

Model	GSM8K	MMLU-Pro
Llama-3.1-8B	+34.0 [28.5, 39.7]	+40.8 [31.8, 50.3]
Mistral-7B-v0.3	+3.5 [−1.2, 8.5]	+8.0 [3.5, 13.2]
Qwen2.5-7B	+4.0 [1.2, 7.3]	+4.8 [1.3, 8.8]
Gemma-3-12B	+2.0 [−0.0, 4.8]	+3.7 [0.2, 7.7]
Qwen3-8B	+0.5 [−1.5, 2.6]	+1.0 [−2.4, 4.8]
Qwen3-4B	+2.1 [0.4, 3.9]	+0.7 [−1.0, 2.5]
Qwen3-1.7B	−0.3 [−3.8, 3.0]	+0.7 [−3.4, 5.0]

at least one task CI excluding zero, and Llama drops 34 to 41 pp. Four controls rule out the obvious confounds. It is not accuracy headroom: Qwen3-1.7B (74% on GSM8K) sits in Llama’s accuracy regime (78%) with the opposite temperature behavior. Model age does not explain it either: Mistral-7B-v0.3 predates Llama-3.1 and is robust. The item subset is not driving it: the stratified MMLU-Pro replication gives Llama +43.3 pp [31.3, 55.3], Qwen2.5 +6.0, and Gemma +4.5. And instruction tuning alone cannot be the answer, since every model here is instruction-tuned (Grover, 2026).

The fragility also runs in the lineage. A follow-up cell repeats the pure-temperature axis on two more Llama generations, Llama-3-8B and Llama-3.2-3B (Q8, same frozen items, $K=3$; Table 7, Appendix D). Llama-3.2-3B collapses like Llama-3.1-8B (+38.0 pp [29.3, 47.3] on GSM8K, +36.7 [26.7, 46.0] on MMLU-Pro) and reproduces the degeneration signature of Section 4.3: the strict-parse rate falls from 99 to 41% (GSM8K) and from 83 to 6% (MMLU-Pro), about half of all generations run to the token cap, and accuracy among well-formed answers stays at or above the greedy level. Llama-3-8B sits between the camps: a null on GSM8K (+4.0 [−2.7, 11.3]) but +16.7 pp [9.3, 24.0] on MMLU-Pro, the longer task, a larger drop than any non-Llama model shows in Table 1. Three Llama generations produce the three largest MMLU-Pro drops in the study, so the fragility is a family trait; what changed in the family’s post-training is the open question of Section 5.

4.2 Sampler choice barely matters before the cliff

At $T=1.3$ the maximum minus minimum accuracy across the seven stochastic sampler arms is 41.8 pp (MMLU-Pro) and 36.7 pp (GSM8K) for Llama, and 1.3 to 10.0 pp for every other model (Figure 2).

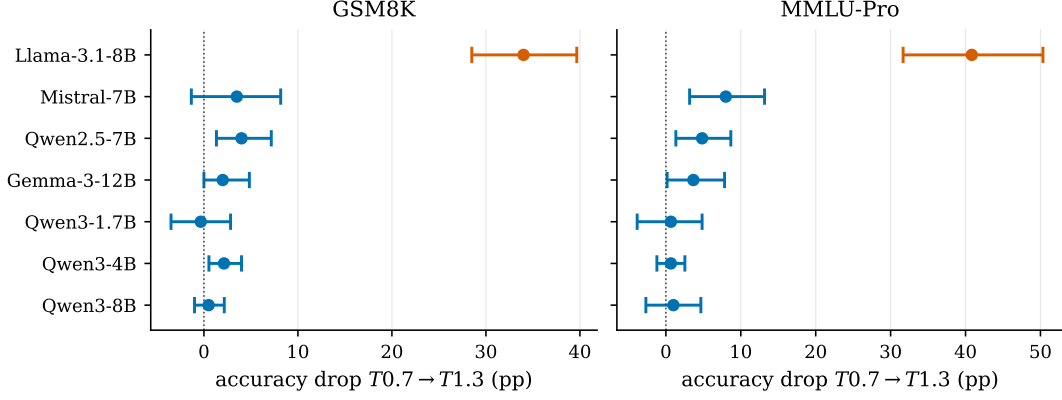


Figure 1: The pure-temperature collapse is model-bound: accuracy drop $T_{0.7} \rightarrow T_{1.3}$, points with 95% item-clustered bootstrap CIs.

The max-minus-min statistic is biased upward under sampling noise (the largest and smallest of several noisy estimates land beyond the largest and smallest true values), so the small spreads are upper bounds; the true differences on robust models are smaller. Chain order (temperature first versus last) is a no-op on robust models ($\pm 1-2$ pp). On Llama at $T=1.3$ it does matter: temperature-last truncation is more protective (top- p 64% $\rightarrow$ 73%, min- p 72% $\rightarrow$ 76% on GSM8K), as expected, because truncating before temperature scaling fixes the surviving token set at its low-temperature size.

4.3 The collapse is degeneration, not wrong answers

We decompose accuracy by parse path (Figure 3). The drop comes almost entirely from generations that never finish: at $T=1.3$, 53.8% (GSM8K) and 89.2% (MMLU-Pro) of Llama generations derail into token-level noise and run to the token cap. Among the generations that do produce a well-formed final answer, Llama still scores 97.3% on GSM8K at $T=1.3$. That number needs one correction before it can be taken at face value: survival is selective, because easier items produce shorter generations and shorter generations have less room to derail. The fair baseline is therefore greedy decoding restricted to the same surviving items, which scores 95.8% (against 76.0% over all items). Conditional on surviving, then, accuracy at $T=1.3$ matches greedy decoding (97.3 vs. 95.8% on GSM8K; 76.9 vs. 78.8% on MMLU-Pro).

The two tasks disguise one failure mode differently. Degenerate text almost always contains a number, so the GSM8K flexible last-number fallback grades it as a wrong answer; the MMLU-Pro

letter parser reports it as a parse failure. The underlying event is the same on both tasks: the generation derails before it reaches an answer.

4.4 The cliff map: robustness is cliff location

We extend the temperature axis to $T \in \{1.3, 1.5, 1.7, 2.0\}$ on three models spanning the gradient (Q8, both tasks, $N=50$, $K=3$; Figure 4). Under plain temperature on GSM8K the three models form a staircase, one rung apart: Llama falls 45 $\rightarrow$ 9 $\rightarrow$ 1 $\rightarrow$ 0%, Qwen2.5 falls 89 $\rightarrow$ 69 $\rightarrow$ 23 $\rightarrow$ 2%, and Gemma falls 89 $\rightarrow$ 77 $\rightarrow$ 49 $\rightarrow$ 8%. The drops against each model’s own $T=0.7$ anchor order the staircase at every rung: at $T=1.5$ Llama (+68.0 pp [58.0, 76.7]) is disjoint from both Qwen2.5 (+24.0 [15.3, 33.3]) and Gemma (+14.7 [8.7, 21.3]), whose intervals still touch; by $T=1.7$ all three are pairwise disjoint (Qwen2.5 +70.0 [62.0, 77.3] versus Gemma +42.7 [34.7, 50.7]). Every model has a cliff; a robust model simply has it later, roughly 0.2 in temperature per step of the gradient. Within each model the cliff sits earlier on MMLU-Pro, the task with roughly twice the generation length; this is what a compounding per-token derail risk predicts. There Llama is at 3% already at $T=1.3$, Qwen2.5 falls 57 $\rightarrow$ 25 $\rightarrow$ 4 $\rightarrow$ 0%, and Gemma falls 60 $\rightarrow$ 24 $\rightarrow$ 6 $\rightarrow$ 0%. The longer task also compresses the staircase: Qwen2.5 and Gemma land on the same MMLU-Pro rung ($T=1.5$) while remaining one rung apart on GSM8K.

Truncation shifts the cliff right in order of tail-cutting strength. Top- p buys about 0.2 in temperature. Min- p holds 70% at $T=1.5$ on Llama GSM8K where plain temperature is at 9%, 85–87% on Qwen2.5 and Gemma GSM8K out to

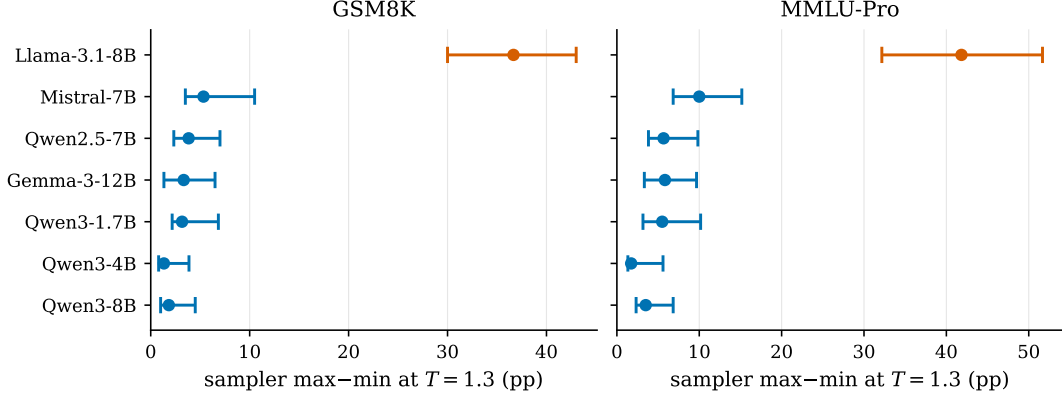


Figure 2: Sampler max–min accuracy spread at $T=1.3$ per model (95% CIs). The statistic is biased upward, so robust-model spreads are upper bounds.

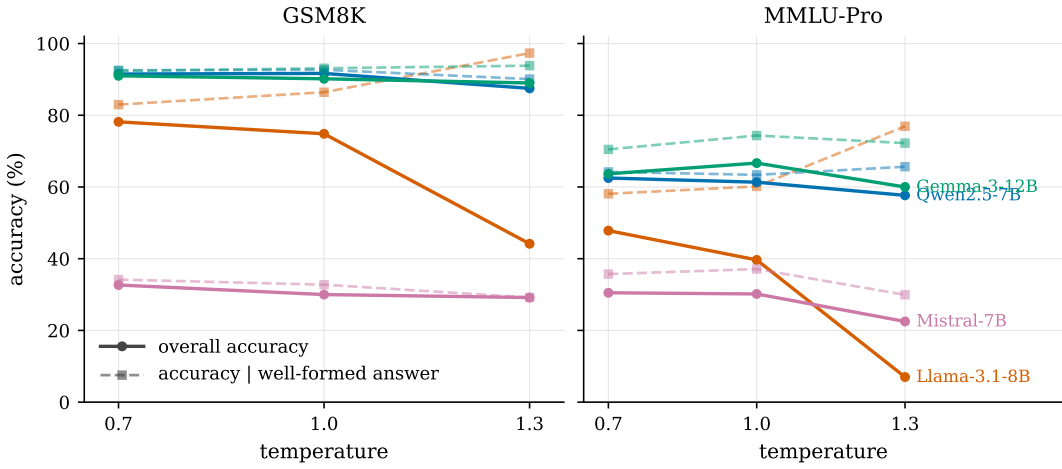


Figure 3: Overall accuracy (solid) versus accuracy among well-formed, strict-parsed answers (dashed) under plain temperature sampling. The dashed lines stay flat: among well-formed answers, accuracy barely moves with temperature.

$T=2.0$, and 53 to 59% on Qwen MMLU-Pro. Top- $n\sigma$ is flat to $T=2.0$ on all three models (75 to 93% on GSM8K; 56 to 66% on Qwen MMLU-Pro), consistent with its construction: the surviving token set is temperature-invariant. The strict-parse rate tracks accuracy almost one-to-one along the whole ladder, so the cliff *is* the degeneration cliff. This reconciles the sampler papers with our null result: min- p and top- $n\sigma$ validate at $T \geq 1.5$ to 3, past every model’s deployment range. There the gains are real and ordered top- $n\sigma \gg$ min- $p \gg$ top- p (the $T=2-3$ probe in Appendix E), but at $T \leq 1.3$ on a robust model the samplers have no failure left to prevent.

4.5 Quantization and self-consistency

Greedy accuracy from Q8_0 to Q3_K_M (pooled tasks) is null on every model of 7B parameters

or more (Gemma $-3.0 [-9, +3]$) and -9 pp on Qwen3-1.7B: the effect grows as models shrink, and nowhere does it reorder samplers (Figure 6). Self-consistency at $T=1.3$: robust models emit three distinct token sequences across $K=3$ repetitions yet converge to 1.1 to 1.4 distinct parsed answers per item; Llama and Mistral diverge to 2.2 to 2.5.

4.6 Toward a mechanism: per-token uncertainty ranks with fragility

If the collapse is a derailment process, fragile models should carry more per-token uncertainty before any derailment happens. We probe this directly: for every model (Q8) we decode the frozen 100 prompts greedily and record the raw top-20 next-token probabilities at each position, summarized per prompt as an entropy lower bound, the top-

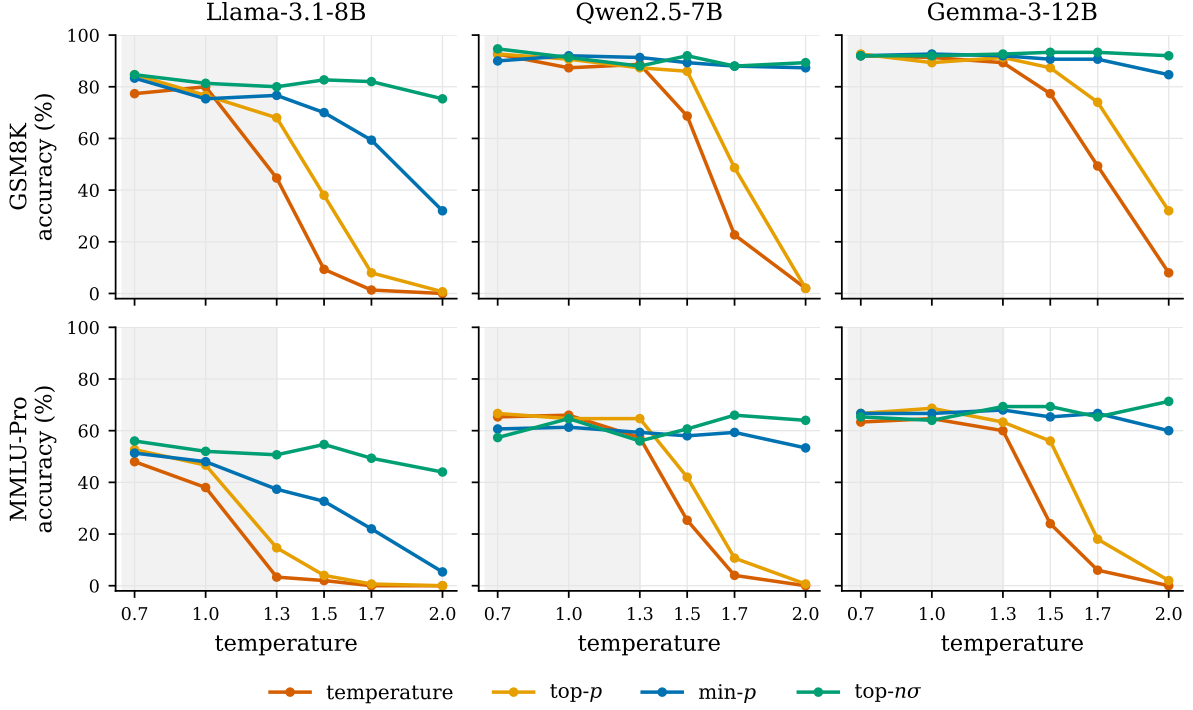


Figure 4: The cliff map. Accuracy versus temperature per sampler (anchors $T \leq 1.0$ from the main matrix, Q8). Every model has a cliff; where it sits is the model property. Truncation shifts the cliff right by tail-cutting strength.

1 probability, the top-1 minus top-2 margin, and the fraction of low-confidence positions (top-1 < 0.5). Greedy decoding holds the trajectory at the model’s modal path, so the measurement is not confounded by sampled derailments.

Mean greedy-path entropy ranks with the observed temperature drop at Spearman $\rho = +0.96$ across the seven models (flat-position fraction $+0.96$; top-1 margin -0.96 ; Figure 5). The ranking is informative and so is its failure: Llama-3.1 (entropy 0.33 nats) and Mistral (0.32) are nearly tied on every top-20 summary, including the mass beyond the top 20 (1.0×10^{-3} both), yet their drops differ five-fold (40.8 versus 8.0 pp).

Generation lengths rule out one candidate mechanism: derail-and-recover. If models derailed and recovered, terminated generations at $T=1.3$ would carry the extra tokens of the detour. They do not: on every robust model the terminated-length distribution is essentially unchanged from $T=0.7$ (90th-percentile ratio 0.85 to 1.19 across models and tasks; Table 4). Llama’s surviving generations at $T=1.3$ are *shorter* than at $T=0.7$ (90th percentile $\times 0.73$ on GSM8K, $\times 0.55$ on MMLU-Pro): exactly the censoring an absorbing failure with a compounding per-token hazard produces, where the generations that survive are the ones

short enough to finish before the first bad draw. Derailment is an absorbing state in every model.

We also estimated the hazard directly. Re-decoding the frozen prompts and storing the raw top-20 arrays at every position, we compute, for each position, the probability that sampling at $T=1.3$ picks a token from beyond the model’s own top 20: 0.0033 per position for Llama versus 0.0034 for Mistral (about 0.85 expected tail draws per 256 positions for both; across all seven models this hazard ranks with the drop at the same $\rho = +0.96$). Both models take the first off-distribution token equally often; only Llama spirals on it. A single absorbed token leaves no length footprint, so this is consistent with the unchanged terminated lengths above. Whatever separates the two models is not visible in the next-token distribution, scaled or unscaled. That leaves one candidate: how the model behaves after an off-distribution token has entered its own context.

A perturbation experiment measures that instability directly. For each model and frozen prompt we greedy-decode a 48-token prefix, force the model’s own rank-20 candidate at that position (median raw probability 2×10^{-9} : a genuine tail draw), and continue greedily for 256 tokens against a control continuation of the unedited pre-

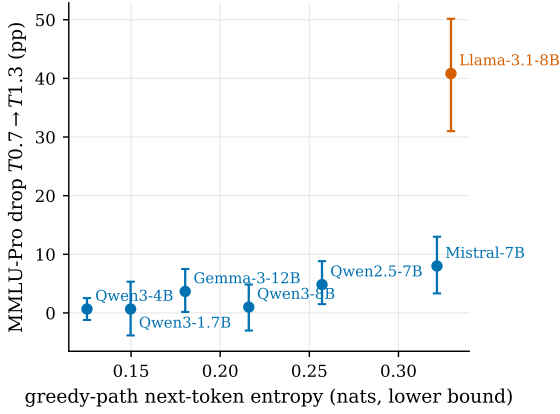


Figure 5: Per-token uncertainty on the model’s own greedy path versus the temperature drop (MMLU-Pro, 95% CIs). The rank correlation is $\rho = +0.96$; the Llama–Mistral gap at equal entropy shows flatness alone does not set the magnitude.

fix. On the surface the forced token is absorbed: continuations stay locally coherent, and Llama’s reach the token cap no more often than its controls (14.0% both, GSM8K). The escape hazard says otherwise. Over the 32 positions after the injection, Llama’s beyond-top-20 escape hazard at $T=1.3$ rises to 2.3 times its baseline (+0.0042 per position); Mistral’s rises 1.35 times (+0.0012), and no robust model gains more than +0.0014. The per-model destabilization ranks with the observed temperature drop at $\rho = +0.86$ (entropy delta: +0.82). This is the compounding loop measured directly: one off-distribution token in context multiplies the probability of drawing the next one, and what separates the fragile model is the multiplier, not the base rate (Appendix I consolidates all per-model probe numbers). With seven models these are consistency checks, not a confirmed causal account, and we report them as such.

5 Discussion

For practitioners. Pick the model, not the sampler. Sampler tuning is insurance against a failure mode that current post-training has largely removed at deployment temperatures.

For the sampler literature. Truncation samplers solve a real but receding problem. Their standard evaluation regime ($T \geq 2$) measures the size of the safety margin, not deployment gains. Top- $n\sigma$ ’s temperature invariance does exactly what it promises (flat to $T=2.0$ on all three ladder models), but the benefit only appears past the cliff.

For the robustness line. Instruction tuning

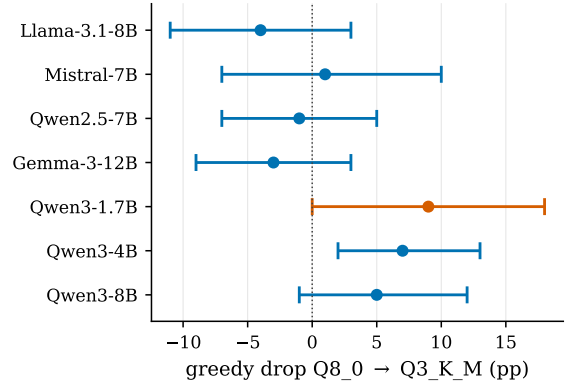


Figure 6: Greedy accuracy drop Q8_0 → Q3_K_M per model (95% CIs), pooled tasks.

does not explain the split: all seven models here are instruction-tuned. What changed between Llama-3.1 and the 2025 cohort (post-training recipe, data, RLHF) is an open question; we contribute the behavioral signature a mechanism study should explain: termination failure with intact answer selection, set off by off-distribution tokens that arrive at the same per-token rate in every model but destabilize only one.

Why degeneration matters beyond benchmarks. An answer-selection failure and a derailment failure have different production consequences (silent errors versus detectable garbage) and different fixes (output validation versus sampler guard rails).

6 Conclusion

Across seven models, five samplers, four quantization levels, and two reasoning tasks, we find that the choice of sampler is a second-order effect at deployment temperatures: on robust models the max-minus-min accuracy spread across samplers is 1.3 to 10.0 points at $T=1.3$, while a single model, Llama-3.1-8B, loses 34 to 41 points under pure temperature on the same grid. The collapse is a termination failure: conditional on producing a well-formed answer, accuracy stays at the greedy level in every cell. Every model has a cliff above its deployment range, and robustness is where that cliff sits. Truncation samplers move the cliff to higher temperatures, but the gains begin above deployment settings. Per-token entropy on the greedy decoding path ranks models by fragility (Spearman $\rho = +0.96$ across the seven models), and a single forced tail token multiplies the fragile model’s subsequent escape hazard 2.3-fold where

robust models absorb it. Together these results argue for screening a model’s temperature fragility before deployment rather than tuning the sampler after it. The open question our data sharpen but cannot answer is what, in post-training, moves the cliff.

Limitations

$N=50$ items per task (CIs are shown throughout; the headline effects are 5 to 10 times their CI widths). 0-shot prompting, English, two tasks. The MMLU-Pro head subset is single-category; the stratified replication is in Appendix C. Every fragile model we observe is a Llama: the lineage cell (Section 4.1) shows the fragility spans three Llama generations, so it is a family trait, but with one fragile family we cannot say how common such families are. Mistral is near floor on GSM8K (32.7% at $T=0.7$), which limits its collapse-detection power there (its MMLU-Pro CI excludes zero). The max-minus-min spread is biased upward, which works against our claim. Accuracy conditional on strict parsing has item survivorship, quantified and normalized in Section 4.3. llama.cpp k-quants only, one quantization recipe by design. Thinking mode off by default (the thinking-on spot-check in Appendix G agrees).

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A Full sampler–temperature grids

Accuracy by sampler and temperature, pooled over quantization levels (Tables 2 and 3).

B Degeneration decomposition

Table 5 decomposes accuracy under plain temperature sampling by parse path: the strict final-answer-sentence rate, the parse-failure rate, the fraction of generations that ran to the token cap (including server-recovered degenerate outputs), and accuracy conditional on a strict parse. Two representative degenerate outputs (Llama-3.1-8B, Q8, plain temperature, $T=1.3$, MMLU-Pro; both ran to the 1024-token cap):

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use the gross profit method of
inventory evaluation. Step 1: Calculate
the gross profit earned during the
year. . .
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Table 2: GSM8K accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	78.0	—	—
	temperature	78.2	74.8	44.2
	top- k	79.8	75.2	58.7
	top- p	81.5	78.0	63.8
	min- p	81.2	78.0	72.0
	top- $n\sigma$	78.8	80.7	80.8
	top- p (temp-last)	80.8	78.7	73.0
	min- p (temp-last)	78.7	81.3	76.2
Mistral-7B	greedy ($T=0$)	31.0	—	—
	temperature	32.7	30.0	29.2
	top- k	31.8	31.7	29.8
	top- p	28.2	31.0	34.2
	min- p	32.3	28.3	28.8
	top- $n\sigma$	29.5	30.7	30.2
	top- p (temp-last)	29.5	31.2	29.5
	min- p (temp-last)	29.5	32.0	30.5
Qwen2.5-7B	greedy ($T=0$)	93.5	—	—
	temperature	91.5	91.7	87.5
	top- k	90.2	88.5	88.7
	top- p	92.2	91.2	90.3
	min- p	90.8	90.5	91.0
	top- $n\sigma$	92.7	91.2	90.7
	top- p (temp-last)	90.8	91.3	90.3
	min- p (temp-last)	92.0	90.7	91.3
Gemma-3-12B	greedy ($T=0$)	92.0	—	—
	temperature	91.0	90.2	89.0
	top- k	91.8	91.7	90.7
	top- p	91.5	90.3	90.7
	min- p	91.5	91.3	92.3
	top- $n\sigma$	91.3	91.5	91.0
	top- p (temp-last)	90.2	91.8	91.5
	min- p (temp-last)	91.2	90.8	90.3
Qwen3-1.7B	greedy ($T=0$)	75.5	—	—
	temperature	73.3	73.7	73.7
	top- k	72.8	72.7	73.3
	top- p	73.8	75.0	72.7
	min- p	72.2	71.7	74.0
	top- $n\sigma$	74.3	73.5	72.3
	top- p (temp-last)	73.0	72.0	71.3
	min- p (temp-last)	74.7	73.3	74.5
Qwen3-4B	greedy ($T=0$)	95.2	—	—
	temperature	95.2	94.0	93.1
	top- k	94.0	94.5	94.1
	top- p	95.5	95.5	93.5
	min- p	95.1	94.7	92.9
	top- $n\sigma$	94.8	93.9	93.9
	top- p (temp-last)	94.8	95.2	94.3
	min- p (temp-last)	94.5	93.9	93.9
Qwen3-8B	greedy ($T=0$)	94.5	—	—
	temperature	95.0	94.2	94.5
	top- k	94.8	94.8	93.5
	top- p	94.0	94.3	93.7
	min- p	94.8	95.3	94.7
	top- $n\sigma$	96.3	94.2	94.3
	top- p (temp-last)	93.8	94.0	94.0
	min- p (temp-last)	96.2	94.7	95.3

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 [opens] To determine the correct type

of research methods, we need to
 understand the characteristics of each
 option: A. Non-probability: This method
 involves selecting. . .

Table 3: MMLU-Pro accuracy (%) by sampler and temperature, pooled over quantization.

Model	Sampler	$T=0.7$	$T=1.0$	$T=1.3$
Llama-3.1-8B	greedy ($T=0$)	54.0	—	—
	temperature	47.8	39.7	7.0
	top- k	51.2	43.3	26.8
	top- p	51.3	46.0	16.0
	min- p	51.2	50.8	39.5
	top- $n\sigma$	51.0	50.5	48.8
	top- p (temp-last)	50.3	46.7	37.5
	min- p (temp-last)	51.2	50.2	43.5
Mistral-7B	greedy ($T=0$)	35.5	—	—
	temperature	30.5	30.2	22.5
	top- k	32.5	32.3	28.3
	top- p	30.2	29.5	25.3
	min- p	33.8	30.7	31.0
	top- $n\sigma$	32.3	31.5	31.5
	top- p (temp-last)	29.2	30.2	32.5
	min- p (temp-last)	33.2	33.3	28.7
Qwen2.5-7B	greedy ($T=0$)	64.0	—	—
	temperature	62.5	61.3	57.7
	top- k	62.5	61.7	57.0
	top- p	65.5	62.8	62.7
	min- p	61.3	63.5	59.7
	top- $n\sigma$	62.0	63.5	61.0
	top- p (temp-last)	63.7	62.7	61.3
	min- p (temp-last)	64.2	63.0	62.3
Gemma-3-12B	greedy ($T=0$)	61.0	—	—
	temperature	63.7	66.7	60.0
	top- k	64.3	62.8	63.7
	top- p	65.7	66.7	63.8
	min- p	65.3	67.0	65.5
	top- $n\sigma$	65.2	65.8	65.8
	top- p (temp-last)	65.8	65.0	65.2
	min- p (temp-last)	65.3	63.7	63.8
Qwen3-1.7B	greedy ($T=0$)	45.5	—	—
	temperature	46.0	45.7	45.3
	top- k	47.5	46.5	45.0
	top- p	47.2	47.0	47.2
	min- p	47.0	47.7	46.2
	top- $n\sigma$	46.8	47.2	46.2
	top- p (temp-last)	47.7	46.8	43.5
	min- p (temp-last)	45.7	46.2	49.0
Qwen3-4B	greedy ($T=0$)	61.6	—	—
	temperature	61.9	62.5	61.2
	top- k	63.5	61.1	61.2
	top- p	62.8	61.3	62.8
	min- p	60.9	62.7	62.9
	top- $n\sigma$	62.4	62.4	61.5
	top- p (temp-last)	62.9	62.7	61.7
	min- p (temp-last)	61.3	63.6	61.5
Qwen3-8B	greedy ($T=0$)	60.5	—	—
	temperature	60.5	59.0	59.5
	top- k	60.2	61.3	57.0
	top- p	60.5	61.3	60.5
	min- p	61.0	59.2	59.7
	top- $n\sigma$	60.0	58.8	58.7
	top- p (temp-last)	60.7	59.5	60.5
	min- p (temp-last)	61.3	60.7	59.3

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The pattern is consistent: a coherent opening, a
mid-generation derailment, then token noise until

Table 4: Terminated-length censoring under plain temperature at $T=0.7$ versus $T=1.3$ (pooled over quantization). “At cap” includes server-recovered degenerate streams.

Model	Task	at cap (%)		p90 length		ratio
		0.7	1.3	0.7	1.3	
Llama-3.1-8B	GSM8K	5.2	53.8	318	232	$0.73\times$
	MMLU-Pro	11.0	89.2	633	345	$0.55\times$
Mistral-7B	GSM8K	3.5	8.3	367	378	$1.03\times$
	MMLU-Pro	1.0	12.5	628	750	$1.19\times$
Qwen2.5-7B	GSM8K	0.2	1.7	341	361	$1.06\times$
	MMLU-Pro	2.5	10.2	771	732	$0.95\times$
Gemma-3-12B	GSM8K	2.0	5.3	363	344	$0.95\times$
	MMLU-Pro	9.8	17.2	753	763	$1.01\times$
Qwen3-1.7B	GSM8K	2.0	2.2	324	339	$1.05\times$
	MMLU-Pro	15.8	16.3	779	806	$1.03\times$
Qwen3-4B	GSM8K	0.9	0.8	353	361	$1.02\times$
	MMLU-Pro	11.9	12.4	778	731	$0.94\times$
Qwen3-8B	GSM8K	2.3	2.3	379	389	$1.03\times$
	MMLU-Pro	12.3	16.0	850	724	$0.85\times$

the cap. The model never reaches an answer at all.

Table 4 gives the length-censoring evidence used in Section 4.6: run-to-cap rates and the 90th percentile of completion length among *terminated* generations at $T=0.7$ versus $T=1.3$. Robust models are unchanged (ratio 0.85 to 1.19); Llama’s surviving generations shorten ($\times 0.73$ GSM8K, $\times 0.55$ MMLU-Pro), so derailment is absorbing and survival is decided by whether the generation finishes before the first bad draw.

C Stratified MMLU-Pro replication

The frozen $N=50$ MMLU-Pro head subset is single-category (business) because the test split is sorted by category. Table 6 repeats the pure-temperature axis (Q8, $K=3$) on a deterministic stratified subset with 3–4 items from each of the 14 categories. The collapse ordering reproduces.

D Llama lineage cell

The lineage cell repeats the pure-temperature axis (greedy anchor plus plain temperature at $T \in \{0.7, 1.0, 1.3\}$) on Llama-3-8B and Llama-3.2-3B: Q8_0, both frozen $N=50$ tasks, $K=3$, 2,000 generations, same pipeline, prompts, and seed derivation as the main matrix. Table 7 reports both models next to the Q8-only Llama-3.1-8B slice of the main matrix, which is why the Llama-3.1 numbers differ slightly from the quantization-pooled Table 1. Greedy anchors are ordinary (Llama-3-

8B 76.0% GSM8K / 40.0% MMLU-Pro; Llama-3.2-3B 74.0 / 42.0%). Llama-3.2-3B reproduces the full degeneration signature at $T=1.3$: strict-parse rate $98.7 \rightarrow 41.3\%$ on GSM8K and $82.7 \rightarrow 6.0\%$ on MMLU-Pro, 50 to 51% of generations at the token cap, and accuracy conditional on a strict parse at or above the greedy level (85.5 and 55.6%). Llama-3-8B’s MMLU-Pro drop has a milder surface: strict-parse $97.3 \rightarrow 65.3\%$ with only 4.7% at the cap, so its lost generations mostly terminate as parse failures short of the cap.

E High-temperature probe ($T=2-3$)

A small probe ($N=15$, $K=3$, Q8) in the regime where min- p and top- $n\sigma$ report their largest gains (Table 8). The sampler ordering is top- $n\sigma \gg \min-p \gg \text{top-}p \approx \text{temperature}$; the zero cells are entirely degeneration-to-cap, and even the most robust model collapses under plain temperature.

F Temperature-first versus temperature-last

Both chain orders for the two probability-space truncators (Table 9).

G Thinking-mode spot-check

The matrix runs Qwen3 with thinking off. A confirmatory spot-check (Qwen3-8B, Q8, thinking on, 450 generations, $N=25$, $K=2$) shows the same temperature robustness: under plain temperature, GSM8K 94% at $T=0.7$ versus 96% at $T=1.3$, and MMLU-Pro 56% versus 46% (small sample), with min- p flat on both tasks. The robustness is not an artifact of suppressing the reasoning mode.

H Self-consistency

Mean distinct parsed answers per item across the $K=3$ repetitions under plain temperature (Table 10). Robust models sample diverse token sequences that converge on the same answer; a parse failure counts as one distinct answer, so the statistic is a divergence proxy.

I Mechanism probes

Table 11 consolidates the per-model mechanism evidence of Section 4.6: greedy-path entropy lower bound and flat-position fraction (top-1 < 0.5) from the summary probe; the temperature-scaled beyond-top-20 escape hazard per position

Table 5: Decomposition of the temperature effect (plain temperature, pooled over quantization; %).

Model	Task	T	acc	strict	failed	at cap	acc strict
Llama-3.1-8B	GSM8K	0.7	78.2	93.0	0.0	5.2	83.0
	GSM8K	1.0	74.8	84.7	0.0	7.8	86.4
	GSM8K	1.3	44.2	43.8	0.5	53.8	97.3
	MMLU-Pro	0.7	47.8	82.3	15.3	11.0	58.1
	MMLU-Pro	1.0	39.7	64.8	29.0	23.7	60.2
	MMLU-Pro	1.3	7.0	8.7	84.5	89.2	76.9
Mistral-7B	GSM8K	0.7	32.7	72.2	0.3	3.5	34.2
	GSM8K	1.0	30.0	67.7	0.0	4.3	32.8
	GSM8K	1.3	29.2	62.8	0.0	8.3	29.2
	MMLU-Pro	0.7	30.5	84.0	12.8	1.0	35.7
	MMLU-Pro	1.0	30.2	80.8	17.0	1.0	37.1
	MMLU-Pro	1.3	22.5	73.5	21.2	12.5	29.9
Qwen2.5-7B	GSM8K	0.7	91.5	98.3	0.0	0.2	92.5
	GSM8K	1.0	91.7	95.7	0.0	0.5	92.7
	GSM8K	1.3	87.5	92.5	0.0	1.7	90.1
	MMLU-Pro	0.7	62.5	97.3	2.3	2.5	64.2
	MMLU-Pro	1.0	61.3	96.5	2.5	3.0	63.4
	MMLU-Pro	1.3	57.7	87.3	10.5	10.2	65.6
Gemma-3-12B	GSM8K	0.7	91.0	98.0	0.0	2.0	92.3
	GSM8K	1.0	90.2	96.8	0.0	3.3	93.1
	GSM8K	1.3	89.0	94.5	0.0	5.3	93.8
	MMLU-Pro	0.7	63.7	90.3	9.0	9.8	70.5
	MMLU-Pro	1.0	66.7	89.7	8.2	10.3	74.3
	MMLU-Pro	1.3	60.0	82.8	14.5	17.2	72.2
Qwen3-1.7B	GSM8K	0.7	73.3	20.3	0.0	2.0	73.8
	GSM8K	1.0	73.7	20.0	0.0	0.7	71.7
	GSM8K	1.3	73.7	19.0	0.0	2.2	71.1
	MMLU-Pro	0.7	46.0	83.7	13.3	15.8	54.8
	MMLU-Pro	1.0	45.7	82.2	15.5	16.3	55.4
	MMLU-Pro	1.3	45.3	83.2	13.5	16.3	54.5
Qwen3-4B	GSM8K	0.7	95.2	94.3	0.0	0.9	95.3
	GSM8K	1.0	94.0	92.3	0.0	0.8	94.8
	GSM8K	1.3	93.1	95.6	0.0	0.8	93.7
	MMLU-Pro	0.7	61.9	87.2	12.5	11.9	70.9
	MMLU-Pro	1.0	62.5	87.3	12.0	10.8	71.6
	MMLU-Pro	1.3	61.2	87.5	11.1	12.4	70.0
Qwen3-8B	GSM8K	0.7	95.0	48.2	0.0	2.3	98.3
	GSM8K	1.0	94.2	49.2	0.0	2.3	98.6
	GSM8K	1.3	94.5	52.2	0.0	2.3	98.1
	MMLU-Pro	0.7	60.5	82.5	15.2	12.3	72.3
	MMLU-Pro	1.0	59.0	80.5	15.5	14.8	72.3
	MMLU-Pro	1.3	59.5	79.8	16.3	16.0	73.5

Table 6: Stratified versus head-of-split (business) MMLU-Pro subset; plain temperature accuracy (%) at Q8_0.

Model	Subset	T			
		0.7	1.0	1.3	drop
Llama-3.1-8B	stratified	44.7	38.7	1.3	+43.3 pp
Llama-3.1-8B	head-50	48.0	38.0	10.0	+38.0 pp
Qwen2.5-7B	stratified	54.7	53.3	48.7	+6.0 pp
Qwen2.5-7B	head-50	65.3	66.0	58.0	+7.3 pp
Gemma-3-12B	stratified	64.0	62.7	56.7	+7.3 pp
Gemma-3-12B	head-50	63.3	64.7	60.0	+3.3 pp

at $T=1.3$ from the raw top-20 arrays; and the perturbation deltas (post-injection minus control over 32 positions, escape hazard and entropy) from the forced-token experiment. Escape columns are $\times 10^{-3}$. Every static column ties Llama-3.1 with Mistral; only the perturbation response separates them.

J Reproducibility

- **Inference stack.** llama.cpp at commit 5aba5364 (server build 9456), native /completion endpoint, CUDA builds of the same commit per GPU architecture (sm_75

Table 7: Llama lineage: plain-temperature accuracy and $T_{0.7} \rightarrow 1.3$ drop (pp, 95% item-clustered bootstrap CIs), all at Q8_0. The Llama-3.1-8B rows are the Q8-only slice of the main matrix.

Model	Task	acc (%) at T		
		0.7	1.3	drop [95% CI]
Llama-3-8B	GSM8K	75.3	71.3	+4.0 [−2.7, 11.3]
	MMLU-Pro	44.0	27.3	+16.7 [9.3, 24.0]
Llama-3.1-8B	GSM8K	77.3	47.3	+30.0 [21.3, 38.7]
	MMLU-Pro	48.0	10.0	+38.0 [28.0, 48.7]
Llama-3.2-3B	GSM8K	74.0	36.0	+38.0 [29.3, 47.3]
	MMLU-Pro	40.0	3.3	+36.7 [26.7, 46.0]

- **Code and data.** [TODO: anonymized repository link for submission.]

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for RTX 2080 Ti, sm_120 for RTX 5070 Ti). Server-side BOS insertion is disabled; the client-rendered chat template owns the special tokens, and each record stores the SHA-256 of the exact rendered prompt.

- **Quantizations.** Every quantized checkpoint is a Bartowski imatrix GGUF (one provider, one recipe, so the quantization axis is not confounded by recipe). SHA-256 hashes of all 29 model files are pinned in the repository (DOWNLOADS.md and per-file .meta.json sidecars).
- **Decoding.** Exactly one truncation parameter per request; the sampler chain order is pinned per arm and recorded in every record. Sampler values: top- p 0.95, top- k 40, min- p 0.05, top- $n\sigma$ 1.0 (each method’s recommended value).
- **Seeding.** Every generation’s seed is derived deterministically from (global seed 0, model, quantization, sampler, temperature, task, item, repetition). Greedy decoding is the deterministic $T=0$ anchor with $K=1$.
- **Tasks.** GSM8K (openai/gsm8k, test split, first 50 items) and MMLU-Pro (TIGER-Lab/MMLU-Pro, test split, first 50 items; stratified replication in Appendix C). Parsers mirror lm-eval-harness; parse failures grade as incorrect and are reported.
- **Records.** Each generation is one JSONL record carrying the full identity tuple, the request parameters, the raw output, the parse path, and throughput covariates; runs are resumable by record id. Batched serving (≤ 8 slots) is seed-logged but not bit-exact.

Table 8: Accuracy (%) at $T=2$ and $T=3$ (45 observations per cell).

Task	Sampler	Llama-3.1-8B		Gemma-3-12B	
		$T=2$	$T=3$	$T=2$	$T=3$
GSM8K	temperature	0	0	16	0
	top- p	0	0	27	0
	min- p	33	0	78	24
	top- $n\sigma$	71	80	87	89
MMLU-Pro	temperature	0	0	0	0
	top- p	0	0	0	0
	min- p	0	0	58	0
	top- $n\sigma$	47	51	69	62

Table 9: Accuracy (%) with temperature applied before (left) versus after (right) truncation, pooled over quantization.

Model	Method	temperature-first			temperature-last		
		0.7	1.0	1.3	0.7	1.0	1.3
<i>GSM8K</i>							
Llama-3.1-8B	top- <i>p</i>	81.5	78.0	63.8	80.8	78.7	73.0
Llama-3.1-8B	min- <i>p</i>	81.2	78.0	72.0	78.7	81.3	76.2
Mistral-7B	top- <i>p</i>	28.2	31.0	34.2	29.5	31.2	29.5
Mistral-7B	min- <i>p</i>	32.3	28.3	28.8	29.5	32.0	30.5
Qwen2.5-7B	top- <i>p</i>	92.2	91.2	90.3	90.8	91.3	90.3
Qwen2.5-7B	min- <i>p</i>	90.8	90.5	91.0	92.0	90.7	91.3
Gemma-3-12B	top- <i>p</i>	91.5	90.3	90.7	90.2	91.8	91.5
Gemma-3-12B	min- <i>p</i>	91.5	91.3	92.3	91.2	90.8	90.3
Qwen3-1.7B	top- <i>p</i>	73.8	75.0	72.7	73.0	72.0	71.3
Qwen3-1.7B	min- <i>p</i>	72.2	71.7	74.0	74.7	73.3	74.5
Qwen3-4B	top- <i>p</i>	95.5	95.5	93.5	94.8	95.2	94.3
Qwen3-4B	min- <i>p</i>	95.1	94.7	92.9	94.5	93.9	93.9
Qwen3-8B	top- <i>p</i>	94.0	94.3	93.7	93.8	94.0	94.0
Qwen3-8B	min- <i>p</i>	94.8	95.3	94.7	96.2	94.7	95.3
<i>MMLU-Pro</i>							
Llama-3.1-8B	top- <i>p</i>	51.3	46.0	16.0	50.3	46.7	37.5
Llama-3.1-8B	min- <i>p</i>	51.2	50.8	39.5	51.2	50.2	43.5
Mistral-7B	top- <i>p</i>	30.2	29.5	25.3	29.2	30.2	32.5
Mistral-7B	min- <i>p</i>	33.8	30.7	31.0	33.2	33.3	28.7
Qwen2.5-7B	top- <i>p</i>	65.5	62.8	62.7	63.7	62.7	61.3
Qwen2.5-7B	min- <i>p</i>	61.3	63.5	59.7	64.2	63.0	62.3
Gemma-3-12B	top- <i>p</i>	65.7	66.7	63.8	65.8	65.0	65.2
Gemma-3-12B	min- <i>p</i>	65.3	67.0	65.5	65.3	63.7	63.8
Qwen3-1.7B	top- <i>p</i>	47.2	47.0	47.2	47.7	46.8	43.5
Qwen3-1.7B	min- <i>p</i>	47.0	47.7	46.2	45.7	46.2	49.0
Qwen3-4B	top- <i>p</i>	62.8	61.3	62.8	62.9	62.7	61.7
Qwen3-4B	min- <i>p</i>	60.9	62.7	62.9	61.3	63.6	61.5
Qwen3-8B	top- <i>p</i>	60.5	61.3	60.5	60.7	59.5	60.5
Qwen3-8B	min- <i>p</i>	61.0	59.2	59.7	61.3	60.7	59.3

Table 10: Mean distinct parsed answers per item ($K=3$, pooled over quantization).

Model	GSM8K		MMLU-Pro	
	$T=0.7$	$T=1.3$	$T=0.7$	$T=1.3$
Llama-3.1-8B	1.51	2.33	1.71	1.28
Mistral-7B	2.21	2.48	1.97	2.21
Qwen2.5-7B	1.10	1.27	1.44	1.57
Gemma-3-12B	1.12	1.16	1.33	1.45
Qwen3-1.7B	1.41	1.43	1.67	1.73
Qwen3-4B	1.05	1.14	1.33	1.38
Qwen3-8B	1.09	1.13	1.41	1.40

Table 11: Mechanism probes per model (Q8, 100 frozen prompts). Escape hazards are at $T=1.3$; drop is the MMLU-Pro plain-temperature $T_{0.7} \rightarrow 1.3$ accuracy drop (pp), pooled over quantization.

Model	H_{lb}	flat%	esc	Δ_{esc}	ΔH	drop
Llama-3.1-8B	0.329	7.1	3.26	+4.20	+0.110	+40.8
Mistral-7B	0.322	6.8	3.40	+1.19	+0.044	+8.0
Qwen2.5-7B	0.257	4.8	1.97	+1.00	+0.049	+4.8
Gemma-3-12B	0.180	3.0	1.44	+1.42	+0.055	+3.7
Qwen3-1.7B	0.150	2.1	0.36	-0.00	+0.012	+0.7
Qwen3-4B	0.125	1.5	0.22	+0.21	+0.021	+0.7
Qwen3-8B	0.216	3.8	0.71	+0.22	+0.024	+1.0